Year 12 Maths - Logarithms

Similar questions - Laws of logarithms and equations

Two new questions on this topic, a little more open-ended than the exam questions. They are not past exam questions. Answers are at the end.

1

(a) Solve $\log_3(x+2)+\log_3(x-4)=3$, explaining why one root of your quadratic must be rejected. **[5]**

(b) Solve $\log_2(3y)-\log_2(y-1)=2$. **[3]**

2

(a) Given that $\log_a 5 = p$ and $\log_a 3 = q$, express $\log_a 75$ and $\log_a\left(\frac{5}{9}\right)$ in terms of p and q. **[3]**

(b) Show that $9^x - 10\times 3^x + 9 = 0$ can be written as a quadratic equation in 3^x, and hence solve it. **[4]**

(c) Using logarithms, solve $5^{x+1} = 2^{3x}$, giving your answer correct to 3 significant figures. **[3]**

Question 1

(a) $\log_3\left[(x+2)(x-4)\right]=3$, so $(x+2)(x-4)=27$ and $x^2-2x-35=0$, giving $(x-7)(x+5)=0$. $x=-5$ must be rejected because $\log_3(x-4)$ would need the logarithm of a negative number. $x=7$

(b) $\log_2\dfrac{3y}{y-1}=2$, so $\dfrac{3y}{y-1}=4$, $3y=4y-4$ and $y=4$

Question 2

(a) $\log_a 75=\log_a(5^2\times 3)=2p+q$. $\log_a\dfrac{5}{9}=p-2q$

(b) $9^x=(3^x)^2$, so with $u=3^x$: $u^2-10u+9=0$, $(u-1)(u-9)=0$. $3^x=1$ gives $x=0$; $3^x=9$ gives $x=2$

(c) $(x+1)\log 5=3x\log 2$, so $x(3\log 2-\log 5)=\log 5$ and $x=\dfrac{\log 5}{3\log 2-\log 5}=3.42$